

RET fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes RET gene fusions from the msk_impact_50k_2026 study, covering 194 fusion events. 146/194 fusions (75.3%) were in-frame. 179/194 fusions (92.3%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.000419666$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 75.3% in-frame and 92.3% domain retention.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 194 analyzed fusion events, identifying 33 distinct fusion partner genes. The most frequent partner was KIF5B, observed in 87 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 141/146 (96.6%) of in-frame fusions retained the domain, $p=0.000419666$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

1.3 Domain disruption

Domain-disruption analysis was skipped: RET has no disruption_required_domains configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 194 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 627 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is -97 aa from the nearest configured domain boundary at 724 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: RET has no group_field configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

1.8 Joint partner

Joint partner reported: RET has no gene_pair configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIF5B	87	0.4484536082474227	0.20043213737826426		0.8093693978698724		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.4372136092846768	1
CCDC6	51	0.26288659793814434	0.20043213737826426		0.8167835941716136		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.35927364693107344	2
PDCD10	1	0.005154639175257732	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2880777515673477	3
TFG	1	0.005154639175257732	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2880777515673477	4
ERC1	1	0.005154639175257732	0.20043213737826426		0.9724770642201834		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28217997961452984	5
NCOA4	16	0.08247422680412371	0.20043213737826426		0.8031823394495413		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27903950471890626	6
RBPM5	1	0.005154639175257732	0.20043213737826426		0.9311926605504587		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27333332168530317	7
RUFY1	2	0.010309278350515464	0.20043213737826426		0.9174311926605505		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2725935667840047	8
LIN52	1	0.005154639175257732	0.20043213737826426		0.908256880733945		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2684185117246216	9
EML4	1	0.005154639175257732	0.20043213737826426		0.8944954128440367		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2654696257482127	10
DOCK1	1	0.005154639175257732	0.20043213737826426		0.8371559633027523		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2531826008465089	11
SPECC1L	4	0.020618556701030927	0.20043213737826426		0.8044724770642202		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.25280638987786913	12
TIMM23B	3	0.015463917525773196	0.20043213737826426		0.8035168195718654		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2503924750543969	13
KCNMA1	1	0.005154639175257732	0.20043213737826426		0.823394495412844		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2502337148701	14
ERCC6	3	0.015463917525773196	0.20043213737826426		0.8027522935779816		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.25022864805570755	15
RUFY3	2	0.010309278350515464	0.20043213737826426		0.805045871559633		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2485109979766652	16
CCDC186	1	0.005154639175257732	0.20043213737826426		0.805045871559633		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24630186690155476	17
CUBN	1	0.005154639175257732	0.20043213737826426		0.805045871559633		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24630186690155476	18
GEMIN5	1	0.005154639175257732	0.20043213737826426		0.805045871559633		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24630186690155476	19
KIF13A	1	0.005154639175257732	0.20043213737826426		0.805045871559633		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24630186690155476	20
RELCH	1	0.005154639175257732	0.20043213737826426		0.805045871559633		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24630186690155476	21
SLC4A4	1	0.005154639175257732	0.20043213737826426		0.805045871559633		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24630186690155476	22
CLIP1	1	0.005154639175257732	0.20043213737826426		0.8027522935779816		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24581038590548662	23
GRIPAP1	1	0.005154639175257732	0.20043213737826426		0.8027522935779816		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24581038590548662	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
RASSF4	1	0.005154639175257732	0.20043213737826426		0.8027522935779816		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24581038590548662	25
RUFY2	1	0.005154639175257732	0.20043213737826426		0.8027522935779816		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24581038590548662	26
SHROOM3	1	0.005154639175257732	0.20043213737826426		0.8027522935779816		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24581038590548662	27
TRIM24	1	0.005154639175257732	0.20043213737826426		0.8027522935779816		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24581038590548662	28
KIAA1217	1	0.005154639175257732	0.20043213737826426		0.536697247706422		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.188798590361581	29
CTNNA3	1	0.005154639175257732	0.20043213737826426		0.29587155963302747		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.13719308577442502	30
CTNNA1	1	0.005154639175257732	0.20043213737826426		0.1009174311926605		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.0954172011086321	31
UXS1	1	0.005154639175257732	0.20043213737826426		0.06422018348623848		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08755350517154166	32
FBXL7	2	0.010309278350515464	0.20043213737826426		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07600116835674388	33

2.2 cutpoint_detection cutpoint_scan

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cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
219	0	179	1	14	0.0	0.07731958762886598	1.1117104708745449
425	1	178	1	14	0.07865168539325842	0.14903050050745154	0.826724840000421
550	3	176	2	13	0.11079545454545454	0.048844025264081076	1.3111885527954676
556	4	175	2	13	0.14857142857142858	0.06998082470265705	1.1550209437950567
587	5	174	3	12	0.11494252873563218	0.016699929087495975	1.7772853729826312
622	6	173	3	12	0.13872832369942195	0.023865412628667555	1.6222310523597865
627	8	171	5	10	0.0935672514619883	0.001234261100135449	2.9085929583309174
639	9	170	5	10	0.10588235294117647	0.0018331517060542683	2.7368015926613842
640	10	169	5	10	0.11834319526627218	0.0026250181738802633	2.5808676854814183
657	11	168	5	10	0.13095238095238096	0.003644511061609513	2.438360727287861
663	13	166	5	10	0.1566265060240964	0.006510395780065948	2.1863926089934114
671	15	164	5	10	0.18292682926829268	0.01072245099455107	1.9697059299659876
673	16	163	5	10	0.19631901840490798	0.01342338349072885	1.8721380023380016
687	17	162	5	10	0.20987654320987653	0.01656697136370287	1.7807568784395744
708	18	161	5	10	0.2236024844720497	0.020185752752359003	1.694955050558838
712	88	91	8	7	0.8461538461538461	0.7940441928844832	0.10015532604674418
713	176	3	14	1	4.190476190476191	0.27712145920070164	0.5573298428514589
934	176	3	15	0	0.0	1.0	-0.0
1019	177	2	15	0	0.0	1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

141	38
5	10

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	194	179	1	14	0.04988179669030733

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
CCDC186	1
CCDC6	51
CLIP1	1
CTNNA1	1
CTNNA3	1
CUBN	1
DOCK1	1
EML4	1
ERC1	1
ERCC6	3
FBXL7	2
GEMIN5	1
GRIPAP1	1
KCNMA1	1
KIAA1217	1
KIF13A	1
KIF5B	87
LIN52	1
NCOA4	16
PDCD10	1
RASSF4	1
RBPMS	1
RELCH	1
RUFY1	2
RUFY2	1
RUFY3	2
SHROOM3	1
SLC4A4	1
SPECC1L	4
TFG	1
TIMM23B	3
TRIM24	1
UXS1	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_ratio	is_breakpoint_in_domain	main_retention_fraction	is_retained_in_domain	disrupted_domain	source_annotation	effect_on_fra
EVT-P-0051781-T03-IM6-3	P-0051781-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610377	11	712	True	retained	1.0	False	Protein kinase domain	CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c2136+193:RET	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	inframe	retained	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0059857-T02-IM7-4	P-0059857-T02-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	43611615	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.453+8003:CCDC6_c.2137-417:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12221:CCDC6_c.2137-334:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-3450:CCDC6_c.2137-327:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv Protein Fusion: out of frame {CCDC6:RET}	NA	
EVT-P-0065992-T01-IM7-5	P-0065992-T01-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	43611698	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12221:CCDC6_c.2137-334:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-3450:CCDC6_c.2137-327:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv Protein Fusion: out of frame {CCDC6:RET}	NA	
EVT-P-0049461-T02-IM6-6	P-0049461-T02-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611705	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-3450:CCDC6_c.2137-327:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv Protein Fusion: out of frame {CCDC6:RET}	NA	
EVT-P-0036379-T01-IM6-7	P-0036379-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611418	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv Protein Fusion: out of frame {CCDC6:RET}	NA	
EVT-P-0037412-T01-IM6-8	P-0037412-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610839	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv Protein Fusion: out of frame {CCDC6:RET}	NA	
EVT-P-0048611-T01-IM6-9	P-0048611-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611237	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv Protein Fusion: out of frame {CCDC6:RET}	NA	
EVT-P-0064853-T01-IM7-10	P-0064853-T01-IM7		CCDC6-RET	CCDC6	out-of-frame	three_prime	43609511	10	627	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv Protein Fusion: out of frame {CCDC6:RET}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_in_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0047144-T01-IM6-11	P-0047144-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611888	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8847:CCDC6_c.2137-144:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6(NM_005436) - RET (NM_020975.4) Fusion (CCDC6 exon1 fused with RET exon12):c.304-2131:CCDC6_c.2136+410:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv Protein Fusion: mid-exon {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv Protein Fusion: in frame {CCDC6:RET}		
EVT-P-0038257-T01-IM6-12	P-0038257-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610594	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6(NM_005436) - RET (NM_020975.4) Fusion (CCDC6 exon1 fused with RET exon12):c.304-2131:CCDC6_c.2136+410:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv Protein Fusion: mid-exon {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv Protein Fusion: in frame {CCDC6:RET}		
EVT-P-0017365-T04-IM6-13	P-0017365-T04-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611464	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv Protein Fusion: mid-exon {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv Protein Fusion: in frame {CCDC6:RET}		
EVT-P-0039548-T01-IM6-14	P-0039548-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610843	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv Protein Fusion: mid-exon {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv Protein Fusion: in frame {CCDC6:RET}		
EVT-P-0040111-T01-IM6-15	P-0040111-T01-IM6		CCDC6-RET	CCDC6	unknown	three_prime	43609967	11	640	False	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv Protein Fusion: mid-exon {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv Protein Fusion: in frame {CCDC6:RET}		
EVT-P-0041192-T01-IM6-16	P-0041192-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610571	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv Protein Fusion: mid-exon {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv Protein Fusion: in frame {CCDC6:RET}		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	inframe	main_ret	disrupted	description	annotation	effect_on_fra
EVT-P-0022642-T04-IM6-17	P-0022642-T04-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_c.2136+193:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	
EVT-P-0022642-T03-IM6-18	P-0022642-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_c.2136+193:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	
EVT-P-0051781-T01-IM6-19	P-0051781-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610377	11	712	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_c.2136+193:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	
EVT-P-0042417-T01-IM6-20	P-0042417-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610397	11	712	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_c.2136+193:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	
EVT-P-0004901-T01-IM5-21	P-0004901-T01-IM5		CCDC6-RET	CCDC6	in-frame	three_prime	43611729	12	713	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_c.2136+193:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	
EVT-P-0022380-T03-IM6-22	P-0022380-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610659	11	712	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 NA (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	
EVT-P-0068500-T01-IM7-23	P-0068500-T01-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	43610322	11	712	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	inframe	main_ref	disrupted	description	annotation	effect_on_fra
EVT-P-0057735-T01-IM6-24	P-0057735-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611887	12	713	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+125:CCDC6_c.2137-145:RETinv Protein Fusion: in frame {CCDC6:RET} CLIP1 (NM_001247997) - RET (NM_020975) rearrangement: t(10;12)(q11.2;q24.31)(chr10:g.4361112		
EVT-P-0038053-T01-IM6-26	P-0038053-T01-IM6		CLIP1-RET	CLIP1	unknown	three_prime	43611121	12	713	True	retained	1.0	False	Protein kinase domain		Fusion: mid-exon {CLIP1:RET} NA Antisense fusion	NA	
EVT-P-0000252-T01-IM3-27	P-0000252-T01-IM3		CUBN-RET	CUBN	unknown	three_prime	43610494	11	712	True	retained	1.0	False	Protein kinase domain		ERC1 (NM_178040) - RET (NM_020975) Rearrangement: t(10;12)(q11.1;p13.32)(chr10:g.43609964::chr12:g.12641		
EVT-P-0038256-T01-IM6-28	P-0038256-T01-IM6		ERC1-RET	ERC1	unknown	three_prime	43609964	11	639	False	retained	1.0	False	Protein kinase domain		Fusion: mid-exon {ERC1:RET} GRIPAP1 (NM_020137) - RET (NM_020975) rearrangement: t(X;10)(p11.23q11.21)(chrX:g.48831008		
EVT-P-0034781-T01-IM6-29	P-0034781-T01-IM6		GRIPAP1-RET	GRIPAP1	in-frame	three_prime	43611853	12	713	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {GRIPAP1:RET} KIAA1217 (NM_019590) - RET (NM_020975) fusion (KIAA1217 exons 1-13 fused to RET exons 7-20): c.2390:KIAA1217_c.1275:RETdel		
EVT-P-0026245-T01-IM6-30	P-0026245-T01-IM6		KIAA1217-RET	KIAA1217	unknown	three_prime	43606666	7	425	False	retained	1.0	False	Protein kinase domain		Fusion: mid-exon {KIAA1217:RET} KIF13A (NM_022113) - RET (NM_020975) fusion: t(6;10)(p22.3;q11.21)(chr6:g.17806558::		
EVT-P-0042153-T01-IM6-31	P-0042153-T01-IM6		KIF13A-RET	KIF13A	in-frame	three_prime	43610510	11	712	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {KIF13A:RET}		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_retained	breakpoint_4	main_ref	clonally_deleted	inframe	disrupted_domain	disrupted_domain	description_annotation	effect_on_fra
EVT-P-0019535-T02-IM6-32	P-0019535-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611913	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-2113:KIF5B_c.2137-119:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET} Note: The NA KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozan- tinib. PMID: 23533264 Protein fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET}	2
EVT-P-0039257-T03-IM7-33	P-0039257-T03-IM7		KIF5B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET} Note: The NA KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozan- tinib. PMID: 23533264 Protein fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET}	2
EVT-P-0009514-T01-IM5-34	P-0009514-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611811	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET} Note: The NA KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozan- tinib. PMID: 23533264 Protein fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET}	2
EVT-P-0055183-T01-IM6-35	P-0055183-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610308	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET}	2

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	domain_ref	domain_1	domain_2	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0014808-T01-IM6-42	P-0014808-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611055	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv Protein fusion: in frame (KIF5B-RET)	NA	
EVT-P-0034357-T01-IM6-43	P-0034357-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611187	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-845:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0014808-T02-IM6-44	P-0014808-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611056	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0033700-T02-IM6-45	P-0033700-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611670	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1454:KIF5B_c.2137-362:RET Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0014808-T03-IM6-46	P-0014808-T03-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611056	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0008289-T01-IM5-47	P-0008289-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611748	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-824:KIF5B_c.2137-284:RETinv Protein fusion: in frame (KIF5B-RET)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	maintained_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0021150-T01-IM6-72	P-0021150-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611591	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:RETinv Protein Fusion: in frame {KIF5B:RET}	NA KIF5B_c.2137-	
EVT-P-0021578-T01-IM6-73	P-0021578-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610809	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:RETinv Protein Fusion: in frame {KIF5B:RET}	NA KIF5B_c.2136+624:RETinv	
EVT-P-0021879-T01-IM6-74	P-0021879-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610699	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:RETinv Protein Fusion: in frame {KIF5B:RET}	NA KIF5B_c.2136+515:RETinv	
EVT-P-0021937-T01-IM6-75	P-0021937-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611555	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:RETinv Protein Fusion: in frame {KIF5B:RET}	NA KIF5B_c.2137-	
EVT-P-0021937-T03-IM6-76	P-0021937-T03-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611555	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:RETinv Protein Fusion: in frame {KIF5B:RET}	NA KIF5B_c.2137-	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	lost_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0003125-T01-IM5-85	P-0003125-T01-IM5		KIF5B-RET	KIF5B	in-frame	five_prime	43611581	12	713	True	lost	0.0	False			Protein kinase domain		NA Protein fusion: in frame (RET-KIF5B)	NA	
EVT-P-0002873-T01-IM3-86	P-0002873-T01-IM3		KIF5B-RET	KIF5B	in-frame	three_prime	43611244	12	713	True	retained	1.0	False		Protein kinase domain			NA Protein fusion: in frame (KIF5B-RET)	NA	
EVT-P-0022866-T01-IM6-87	P-0022866-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611871	12	713	True	retained	1.0	False		Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-767:KIF5B_161:RETinv Protein Fusion: in frame {KIF5B:RET}	NA c.2137-	
EVT-P-0024225-T01-IM6-88	P-0024225-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43609110	10	622	False	retained	1.0	False		Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 10-20): c.2762-430:KIF5B_1866:RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA c.1866:RETinv	
EVT-P-0002694-T02-IM5-89	P-0002694-T02-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611682	12	713	True	retained	1.0	False		Protein kinase domain			NA Protein fusion: in frame (KIF5B-RET)	NA	
EVT-P-0024843-T01-IM6-90	P-0024843-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611733	12	713	True	retained	1.0	False		Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20): c.1726-1339:KIF5B_299:RETinv Protein Fusion: in frame {KIF5B:RET}	NA c.2137-	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	maintained	domain	disrupted_domain	description_annotation	effect_on_fra
EVT-P-0055183-T02-IM6-97	P-0055183-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610308	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET}	
EVT-P-0025838-T01-IM6-98	P-0025838-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610972	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET}	
EVT-P-0025928-T01-IM6-99	P-0025928-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET}	
EVT-P-0025928-T02-IM6-100	P-0025928-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET}	
EVT-P-0036637-T01-IM6-101	P-0036637-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611965	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET}	
EVT-P-0039257-T01-IM6-102	P-0039257-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	clonally_expressed	in_frame	domain	disrupted_domain	description_annotation	status	effect_on_fra
EVT-P-0026693-T01-IM6-103	P-0026693-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610511	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion:c.2544+156:KIF5_c.2136+327:RET	NA	
EVT-P-0026777-T01-IM6-104	P-0026777-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610717	11	712	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+538:RET	NA	Inv
EVT-P-0026777-T02-IM6-105	P-0026777-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610717	11	712	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+538:RET	NA	Inv
EVT-P-0038961-T01-IM6-106	P-0038961-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610775	11	712	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-293:KIF5B_c.2136+591:RET	NA	Inv
EVT-P-0027723-T01-IM6-107	P-0027723-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43610171	11	708	False	retained	1.0	False	Protein kinase domain				Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 11-20): c.1725+966:KIF5B_c.2123:RET	NA	Inv

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	fusion_ratio	donor_frame	acceptor_frame	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0036737-T02-IM6-108	P-0036737-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43612025	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1487:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725-286:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B Protein Fusion: out of frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0061331-T01-IM7-109	P-0061331-T01-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43610884	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-286:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B Protein Fusion: out of frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0027991-T01-IM6-110	P-0027991-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43608201	9	550	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B Protein Fusion: out of frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0062948-T03-IM7-111	P-0062948-T03-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43610211	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B Protein Fusion: out of frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0064951-T01-IM7-112	P-0064951-T01-IM7		KIF5B-RET	KIF5B	out-of-frame	three_prime	43611130	12	713	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B Protein Fusion: out of frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0030453-T01-IM6-113	P-0030453-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610911	11	712	True	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B Protein Fusion: in frame {KIF5B:RET}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_retained	donor_frame	main_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0037494-T01-IM6-114	P-0037494-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611755	12	713	True	retained	1.0	False	Protein kinase domain		KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET) RET (NM_020975) Inversion (c.2137-385_KIF5B:c.1726-1971inv) Protein fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET} RET (NM_020975) - LIN52 (NM_001024674) Rear-range-ment : t(10;14)(p32.1;q12)(chr10:g.43608613:ch	NA	
EVT-P-0000208-T02-IM5-115	P-0000208-T02-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611647	12	713	True	retained	1.0	False	Protein kinase domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET) RET (NM_020975) Inversion (c.2137-385_KIF5B:c.1726-1971inv) Protein fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET} RET (NM_020975) - LIN52 (NM_001024674) Rear-range-ment : t(10;14)(p32.1;q12)(chr10:g.43608613:ch	NA	
EVT-P-0000208-T01-IM3-116	P-0000208-T01-IM3		KIF5B-RET	KIF5B	in-frame	three_prime	43611647	12	713	True	retained	1.0	False	Protein kinase domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET) RET (NM_020975) Inversion (c.2137-385_KIF5B:c.1726-1971inv) Protein fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET} RET (NM_020975) - LIN52 (NM_001024674) Rear-range-ment : t(10;14)(p32.1;q12)(chr10:g.43608613:ch	NA	
EVT-P-0016221-T02-IM6-117	P-0016221-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611268	12	713	True	retained	1.0	False	Protein kinase domain		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET} RET (NM_020975) - LIN52 (NM_001024674) Rear-range-ment : t(10;14)(p32.1;q12)(chr10:g.43608613:ch	NA	
EVT-P-0019663-T01-IM6-118	P-0019663-T01-IM6		LIN52-RET	LIN52	out-of-frame	five_prime	43608613	9	587	True	lost	0.0	False	Protein kinase domain		LIN52 (NM_001024674) Rear-range-ment : t(10;14)(p32.1;q12)(chr10:g.43608613:ch	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	clonif	fract	maintained	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0065826-T01-IM7-120	P-0065826-T01-IM7		NCOA4-RET	NCOA4	out-of-frame	three_prime	43610857	11	712	True	retained	1.0	False	Protein kinase domain					NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2136+673:RET_c.*23+937NCOA4dup		
EVT-P-0037961-T01-IM6-121	P-0037961-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611448	12	713	True	retained	1.0	False	Protein kinase domain					{TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.763-360:NCOA4_c.2137-584:RETdup Protein Fusion: out of frame {TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-415:NCOA4_c.2137-51:RETdup Protein Fusion: out of frame {TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2137-188:RET_c.763-448NCOA4dup Protein Fusion: out of frame {TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup Protein Fusion: out of frame {TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-260:NCOA4_c.2137-707:RETdup Protein fusion: in frame (NCOA4-RET)		
EVT-P-0038315-T01-IM6-122	P-0038315-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611981	12	713	True	retained	1.0	False	Protein kinase domain					{TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-415:NCOA4_c.2137-51:RETdup Protein Fusion: out of frame {TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2137-188:RET_c.763-448NCOA4dup Protein Fusion: out of frame {TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup Protein Fusion: out of frame {TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-260:NCOA4_c.2137-707:RETdup Protein fusion: in frame (NCOA4-RET)		
EVT-P-0060765-T02-IM7-123	P-0060765-T02-IM7		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611844	12	713	True	retained	1.0	False	Protein kinase domain					{TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2137-188:RET_c.763-448NCOA4dup Protein Fusion: out of frame {TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup Protein Fusion: out of frame {TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-260:NCOA4_c.2137-707:RETdup Protein fusion: in frame (NCOA4-RET)		
EVT-P-0043077-T01-IM6-124	P-0043077-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611580	12	713	True	retained	1.0	False	Protein kinase domain					{TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup Protein Fusion: out of frame {TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-260:NCOA4_c.2137-707:RETdup Protein fusion: in frame (NCOA4-RET)		
EVT-P-0005748-T01-IM5-125	P-0005748-T01-IM5		NCOA4-RET	NCOA4	in-frame	three_prime	43611325	12	713	True	retained	1.0	False	Protein kinase domain					{TIMM23B} NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-260:NCOA4_c.2137-707:RETdup Protein fusion: in frame (NCOA4-RET)		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1_pos	protein_2_pos	breakpoint_4	main_ref	chr10	fract	inframe	disrupted_domain	disrupted_desc	annotation	effect_on_fra
EVT-P-0047991-T02-IM6-126	P-0047991-T02-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611538	12	713	True	retained	1.0	False	Protein kinase domain				NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+537:NCOA4_c.2137-494:RETdup Protein Fusion: out of frame {TIMM23B:RET} NCOA4 NA (NM_001145260) - RET (NM_020975) fusion: c.851:NCOA4_c.2137-807:RETdup Protein Fusion: mid-exon {NCOA4:RET} NCOA4 NA (NM_001145260) - RET (NM_020975) fusion: c.1889:NCOA4_c.2137-419:RETdup Protein Fusion: mid-exon {NCOA4:RET} NCOA4 NA (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup Protein Fusion: out of frame {TIMM23B:RET} NCOA4 NA (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup Protein Fusion: out of frame {TIMM23B:RET} KIAA1468 NA (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.2;q21.33)(chr10:g.4361027		
EVT-P-0048805-T01-IM6-127	P-0048805-T01-IM6		NCOA4-RET	NCOA4	unknown	three_prime	43611225	12	713	True	retained	1.0	False	Protein kinase domain				NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.851:NCOA4_c.2137-807:RETdup Protein Fusion: mid-exon {NCOA4:RET} NCOA4 NA (NM_001145260) - RET (NM_020975) fusion: c.1889:NCOA4_c.2137-419:RETdup Protein Fusion: mid-exon {NCOA4:RET} NCOA4 NA (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup Protein Fusion: out of frame {TIMM23B:RET} NCOA4 NA (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup Protein Fusion: out of frame {TIMM23B:RET} KIAA1468 NA (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.2;q21.33)(chr10:g.4361027		
EVT-P-0049600-T01-IM6-128	P-0049600-T01-IM6		NCOA4-RET	NCOA4	unknown	three_prime	43611613	12	713	True	retained	1.0	False	Protein kinase domain				NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.1889:NCOA4_c.2137-419:RETdup Protein Fusion: mid-exon {NCOA4:RET} NCOA4 NA (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup Protein Fusion: out of frame {TIMM23B:RET} NCOA4 NA (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup Protein Fusion: out of frame {TIMM23B:RET} KIAA1468 NA (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.2;q21.33)(chr10:g.4361027		
EVT-P-0056870-T01-IM6-129	P-0056870-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43610337	11	712	True	retained	1.0	False	Protein kinase domain				NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup Protein Fusion: out of frame {TIMM23B:RET} NCOA4 NA (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup Protein Fusion: out of frame {TIMM23B:RET} KIAA1468 NA (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.2;q21.33)(chr10:g.4361027		
EVT-P-0052409-T01-IM6-130	P-0052409-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611691	12	713	True	retained	1.0	False	Protein kinase domain				NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup Protein Fusion: out of frame {TIMM23B:RET} KIAA1468 NA (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.2;q21.33)(chr10:g.4361027		
EVT-P-0004094-T02-IM6-131	P-0004094-T02-IM6		RELCH-RET	RELCH	in-frame	three_prime	43610279	11	712	True	retained	1.0	False	Protein kinase domain				RELCH (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.2;q21.33)(chr10:g.4361027		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein_pos	protein_pos	protein_pos	main_retained	donor_inframe	donor_inframe	disrupted_domains	domains_annotation	effect_on_fra	
EVT-P-0015588-T01-IM6-132	P-0015588-T01-IM6		RET-CCDC186	CCDC186	in-frame	three_prime	43610308	11	712	True	retained	1.0	False	Protein kinase domain			C10orf118 (NM_018017) - RET (NM_020975) Rear-range-ment : c.1326+142:Protein fusion: in frame (C10orf118-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.303+2543:848:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - (RET (NM_020975) rearrange-ment: c.304-9458:CCDC6_787:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET ex-ons 12-20): c.303+25501:568:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) inversion: c.304-12339_c.2137-333inv5 Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-5768:CCDC6_872:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	C10orf118_c.2136+124:RET
EVT-P-0020974-T01-IM6-133	P-0020974-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611184	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6_c.2137-		
EVT-P-0001745-T02-IM5-134	P-0001745-T02-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611245	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6_c.2137-		
EVT-P-0017365-T02-IM6-135	P-0017365-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611464	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6_c.2137-		
EVT-P-0001898-T01-IM3-136	P-0001898-T01-IM3		RET-CCDC6	CCDC6	in-frame	three_prime	43611699	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6_c.2137-		
EVT-P-0029602-T01-IM6-137	P-0029602-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611160	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6_c.2137-		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	domain_pos	main_retention	donor_inframe	donor_fused	loss_domains	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0004916-T01-IM5-138	P-0004916-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6(NM_005436)-RET(NM_020975) Fusion(CCDC6 exon 2 fused to RET exon11): c.304-1153:CCDC6_c.2136+169:RETinv Protein fusion: in frame (CCDC6-RET)	NA (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv) Protein fusion: in frame (RET-CCDC6)	CCDC6_c.2136+169:RETinv
EVT-P-0000425-T01-IM3-139	P-0000425-T01-IM3		RET-CCDC6	CCDC6	in-frame	five_prime	43610450	11	712	True	lost	0.0	False	Protein kinase domain	Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv) Protein fusion: in frame (RET-CCDC6)	NA (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv) Protein fusion: in frame (RET-CCDC6)	CCDC6_c.2136+169:RETinv
EVT-P-0031330-T01-IM6-140	P-0031330-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+3703:CCDC6_c.2136+169:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12-20): c.303+10940:CCDC6_c.2137-	CCDC6_c.2136+169:RETinv	
EVT-P-0030990-T03-IM6-141	P-0030990-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611881	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12-20): c.303+10940:CCDC6_c.2137-151:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 Exon1 fused to RET Exon10) : c.304-11493:CCDC6_c.1760-207:RETinv Protein fusion: out of frame (CCDC6-RET)	CCDC6_c.2136+169:RETinv	
EVT-P-0005168-T01-IM5-142	P-0005168-T01-IM5		RET-CCDC6	CCDC6	out-of-frame	three_prime	43608797	10	587	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 Exon1 fused to RET Exon10) : c.304-11493:CCDC6_c.1760-207:RETinv Protein fusion: out of frame (CCDC6-RET)	CCDC6_c.2136+169:RETinv	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	protein_pos	breakpoint_4	main_ref	donor_pos	acceptor_pos	disrupted_domains	domains_annotation	effect_on_fra
EVT-P-0022380-T01-IM6-143	P-0022380-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610659	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:RETinv	
EVT-P-0022380-T02-IM6-144	P-0022380-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610659	11	712	True	retained	1.0	False	Protein kinase domain			{CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:c.2137-38inv	
EVT-P-0029315-T02-IM6-145	P-0029315-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610631	11	712	True	retained	1.0	False	Protein kinase domain			{CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-20462:CCDC6_c.2136+447:RETinv	
EVT-P-0029315-T01-IM6-146	P-0029315-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610631	11	712	True	retained	1.0	False	Protein kinase domain			{CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-20462:CCDC6_c.2136+447:RETinv	
EVT-P-0000889-T01-IM3-147	P-0000889-T01-IM3		RET-CCDC6	CCDC6	in-frame	five_prime	43611994	12	713	True	lost	0.0	False	Protein kinase domain			RET (NM_020630) - CCDC6 (NM_005436) Reciprocal Inversion: c.2137-38_c.2137-38inv Protein fusion: in frame (RET-CCDC6)	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	breakpoint_5	main_ref	donor_frame	inframe	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0007322-T02-IM6-148	P-0007322-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610268	11	712	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion: c.454-1512:CCDC6_c.2136+84:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+20901:CCDC6_c.2137-754:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv Protein fusion: in frame (CCDC6-RET)	NA	
EVT-P-0027953-T01-IM6-149	P-0027953-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611278	12	713	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+20901:CCDC6_c.2137-754:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv Protein fusion: in frame (CCDC6-RET)	NA	
EVT-P-0008504-T03-IM6-150	P-0008504-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610583	11	712	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv Protein fusion: in frame (CCDC6-RET)	NA	
EVT-P-0008955-T03-IM5-151	P-0008955-T03-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611019	11	712	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv Protein fusion: in frame (CCDC6-RET)	NA	
EVT-P-0013023-T01-IM5-152	P-0013023-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611810	12	713	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv Protein fusion: in frame (CCDC6-RET)	NA	
EVT-P-0013394-T01-IM5-153	P-0013394-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43610232	11	712	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv Protein fusion: in frame (CCDC6-RET)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	protein_pos_2	main_retention	donor_pos	truncated	protein_domain	disrupted_domains	annotation	effect_on_fra
EVT-P-0022607-T01-IM6-154	P-0022607-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611212	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12 - 20): c.304-414:CCDC6_c.2137-820:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exon12): c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20): c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv Protein Fusion: in frame {CCDC6:RET}	
EVT-P-0022642-T01-IM6-155	P-0022642-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exon12): c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20): c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 fused to RET exons 12-20): c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 fused to RET exons 12-20): c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	
EVT-P-0022642-T02-IM6-156	P-0022642-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20): c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 fused to RET exons 12-20): c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 fused to RET exons 12-20): c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	
EVT-P-0017365-T03-IM6-157	P-0017365-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611464	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET}	
EVT-P-0025349-T03-IM6-158	P-0025349-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611986	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv Protein Fusion: in frame {CCDC6:RET}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	breakpoint	main_ret	domain	domain	disrupted	description	annotation	effect_on
EVT-P-0049588-T01-IM6-159	P-0049588-T01-IM6		RET-CCDC6	CCDC6	out-of-frame	five_prime	43609437	10	627	True	lost	0.0	False		Protein kinase domain			RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314-15222:CCDC6inv Protein Fusion: out of frame of {RET:CCDC6} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv	NA	
EVT-P-0015435-T01-IM6-160	P-0015435-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610969	11	712	True	retained	1.0	False		Protein kinase domain			{RET:CCDC6} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv	NA	
EVT-P-0005556-T01-IM5-161	P-0005556-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611828	12	713	True	retained	1.0	False		Protein kinase domain			{RET:CCDC6} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 with RET exons 12-20) - c.303+21888:CCDC6_c.2137-204:RETinv Protein Fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: t(5;10)(q31.2;q11.21)(chr5:g.138223827:CCDC6:1461-1461)	NA	
EVT-P-0063502-T01-IM7-162	P-0063502-T01-IM7		RET-CTNNA1	CTNNA1	unknown	five_prime	43622038	19	1019	False	retained	1.0	False		Protein kinase domain			RET (NM_020975) - CTNNA1 (NM_001903) fusion: t(5;10)(q31.2;q11.21)(chr5:g.138223827:CCDC6:1461-1461)	NA	
EVT-P-0063802-T01-IM7-163	P-0063802-T01-IM7		RET-CTNNA3	CTNNA3	out-of-frame	five_prime	43617566	16	934	True	disrupted	0.7482269503461	False		Protein kinase domain			mid-exon {RET:CTNNA1} RET (NM_020975) - CTNNA3 (NM_013246) rearrangement: c.2801+102-34298:CTNNA3inv Protein Fusion: out of frame {RET:CTNNA3} DOCK1 (NM_001380) - RET (NM_020975) Rearrangement: c.2515+295-RET_c.1668:RETdup	NA	
EVT-P-0030396-T02-IM6-164	P-0030396-T02-IM6		RET-DOCK1	DOCK1	unknown	three_prime	43608320	9	556	False	retained	1.0	False		Protein kinase domain			mid-exon {DOCK1:RET}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_positives	protein_negatives	main_retained	donor_inframe	transcript_maintained	domain_deleted	disrupted_domains	annotation	position	effect_on_fusion
EVT-P-0013886-T01-IM5-165	P-0013886-T01-IM5		RET-EML4	EML4	unknown	three_prime	43611067	11	673	False	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - RET (NM_020975) rearrangement: t(2;10)(p21;q11.21)(chr2:g.42550559::chr10:g.111745388::)	NA	
EVT-P-0014808-T03-IM6-166	P-0014808-T03-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv Protein Fusion: out of frame {RET:ERCC6}	NA	
EVT-P-0014808-T02-IM6-167	P-0014808-T02-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv Protein Fusion: out of frame {RET:ERCC6}	NA	
EVT-P-0014808-T01-IM6-168	P-0014808-T01-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv Protein Fusion: out of frame {RET:ERCC6}	NA	
EVT-P-0045809-T02-IM6-169	P-0045809-T02-IM6		RET-FBXL7	FBXL7	in-frame	five_prime	43622269	19	1063	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15;q11.21)(chr5:g.15556734::chr10:g.111745388::)	NA	
EVT-P-0045809-T06-IM6-170	P-0045809-T06-IM6		RET-FBXL7	FBXL7	in-frame	five_prime	43622323	19	1063	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15;q11.21)(chr5:g.15556734::chr10:g.111745388::)	NA	
EVT-P-0025215-T01-IM6-171	P-0025215-T01-IM6		RET-GEMIN5	GEMIN5	in-frame	three_prime	43611067	11	712	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) rearrangement: t(5;10)(q33.2;q11.21)(chr5:g.154274993::chr10:g.111745388::)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	is_disrupted	description	annotation	effect_on_fra
EVT-P-0027417-T01-IM6-173	P-0027417-T01-IM6		RET-KCNMA1	KCNMA1	out-of-frame	three_prime	43608158	9	550	True	retained	1.0	False	Protein kinase domain			KCNMA1 (NM_001161352) - RET (NM_020975) rearrangement: c.378+15627:KCNMA1_c.1649-143:RET Protein Fusion: out of frame of {KC-NMA1:RET} RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.2136+147:RET_c.2545-77:KIF5Binv Protein Fusion: in frame {RET:KIF5B} NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup Protein fusion: in frame (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2137-387_c.1840-75dup Protein fusion: in frame (NCOA4-RET) NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET ex-ons 12=20: c.872:NCOA4_c.2136+621:RETdup Protein fusion: mid-exon (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2136+757_c.715-617dup Protein fusion: in frame (NCOA4-RET)	NA (1352)	
EVT-P-0043704-T01-IM6-174	P-0043704-T01-IM6		RET-KIF5B	KIF5B	in-frame	five_prime	43610331	11	712	True	lost	0.0	False		Protein kinase domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.2136+147:RET_c.2545-77:KIF5Binv Protein Fusion: in frame {RET:KIF5B} NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup Protein fusion: in frame (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2137-387_c.1840-75dup Protein fusion: in frame (NCOA4-RET) NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET ex-ons 12=20: c.872:NCOA4_c.2136+621:RETdup Protein fusion: mid-exon (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2136+757_c.715-617dup Protein fusion: in frame (NCOA4-RET)	NA	
EVT-P-0006226-T02-IM5-178	P-0006226-T02-IM5		RET-NCOA4	NCOA4	in-frame	three_prime	43611893	12	713	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) - NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup Protein fusion: in frame (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2137-387_c.1840-75dup Protein fusion: in frame (NCOA4-RET) NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET ex-ons 12=20: c.872:NCOA4_c.2136+621:RETdup Protein fusion: mid-exon (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2136+757_c.715-617dup Protein fusion: in frame (NCOA4-RET)	NA	
EVT-P-0001151-T01-IM3-179	P-0001151-T01-IM3		RET-NCOA4	NCOA4	in-frame	three_prime	43611645	12	713	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) - NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup Protein fusion: in frame (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2137-387_c.1840-75dup Protein fusion: in frame (NCOA4-RET) NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET ex-ons 12=20: c.872:NCOA4_c.2136+621:RETdup Protein fusion: mid-exon (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2136+757_c.715-617dup Protein fusion: in frame (NCOA4-RET)	NA	
EVT-P-0009592-T01-IM5-180	P-0009592-T01-IM5		RET-NCOA4	NCOA4	unknown	three_prime	43611143	12	713	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) - NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup Protein fusion: in frame (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2137-387_c.1840-75dup Protein fusion: in frame (NCOA4-RET) NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET ex-ons 12=20: c.872:NCOA4_c.2136+621:RETdup Protein fusion: mid-exon (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2136+757_c.715-617dup Protein fusion: in frame (NCOA4-RET)	NA	
EVT-P-0001155-T01-IM3-181	P-0001155-T01-IM3		RET-NCOA4	NCOA4	in-frame	three_prime	43610941	11	712	True	retained	1.0	False	Protein kinase domain			RET (NM_020975) - NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup Protein fusion: in frame (NCOA4-RET) RET (NM_020630) - NCOA4 (NM_005437) Dupli-cation : c.2136+757_c.715-617dup Protein fusion: in frame (NCOA4-RET)	NA	

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3 Figures

Figure 1: domain retention outliers

partner block → breakpoint → retained RET portion (domain-colored, exon ticks)



Figure 2: fusion schematic

Reference vs msk_impact_50k_2026



Figure 3: reference comparison